

Detection recall vs. coverage

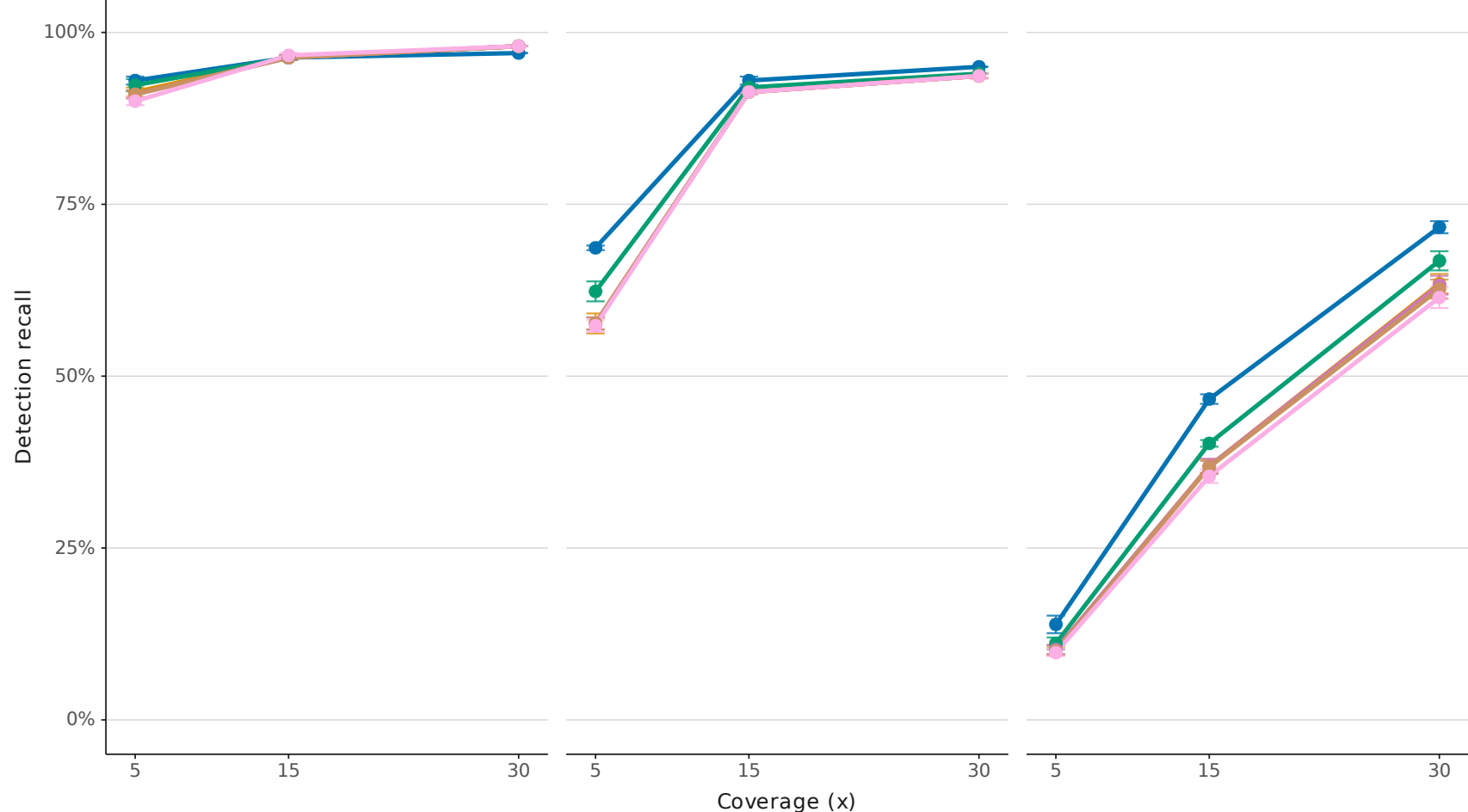
Fraction of simulated insertions recovered (mean $\pm$ SEM, 3 replicates)

- RelocaTE2
- RelocaTE3-blat/bwaaln
- RelocaTE3-bwa/bwa
- RelocaTE3-blat/bwa
- RelocaTE3-bowtie2/bwa
- RelocaTE3-minimap2/minimap2

Homozygous

Heterozygous

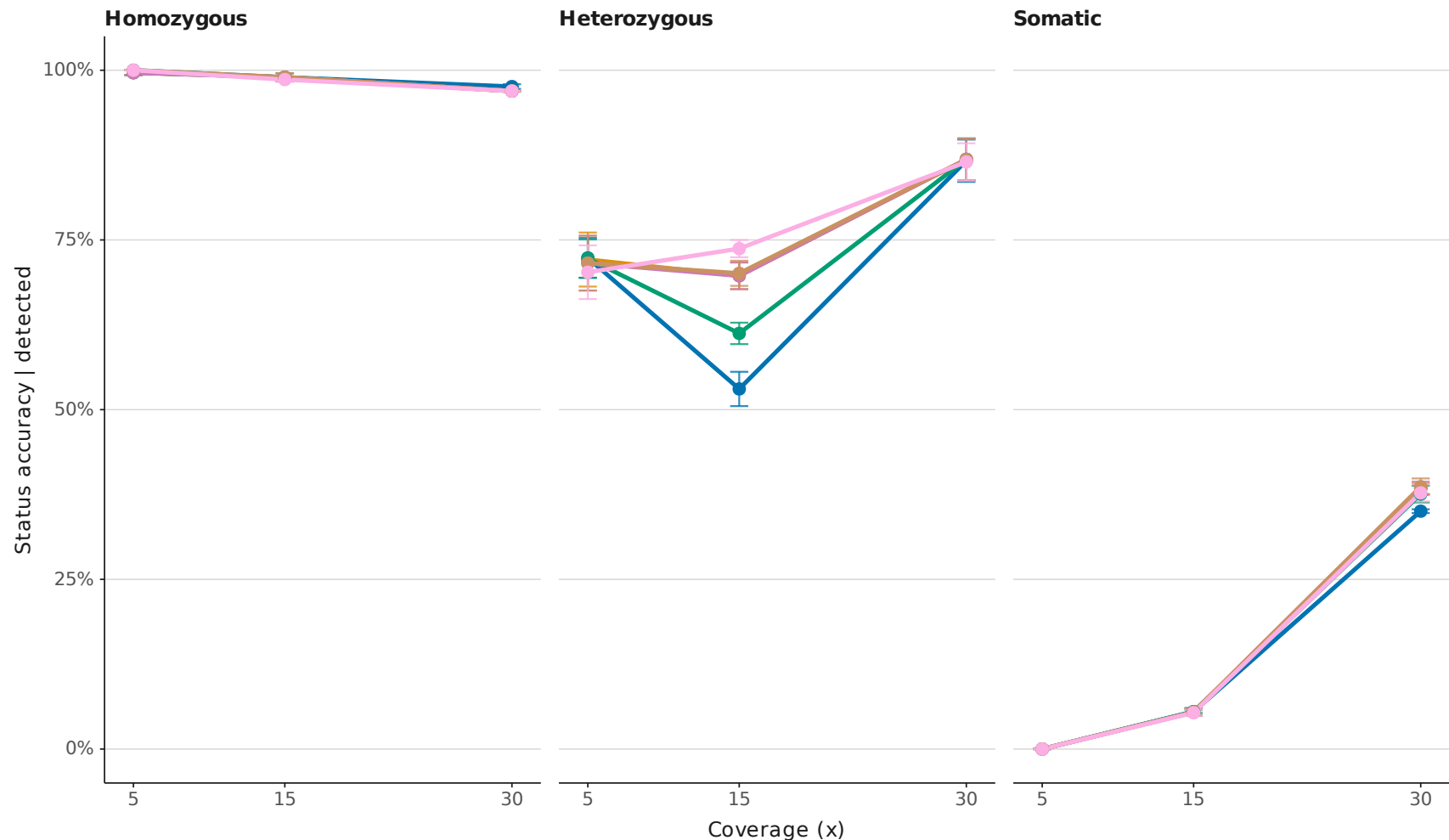
Somatic



Genotype-status accuracy vs. coverage

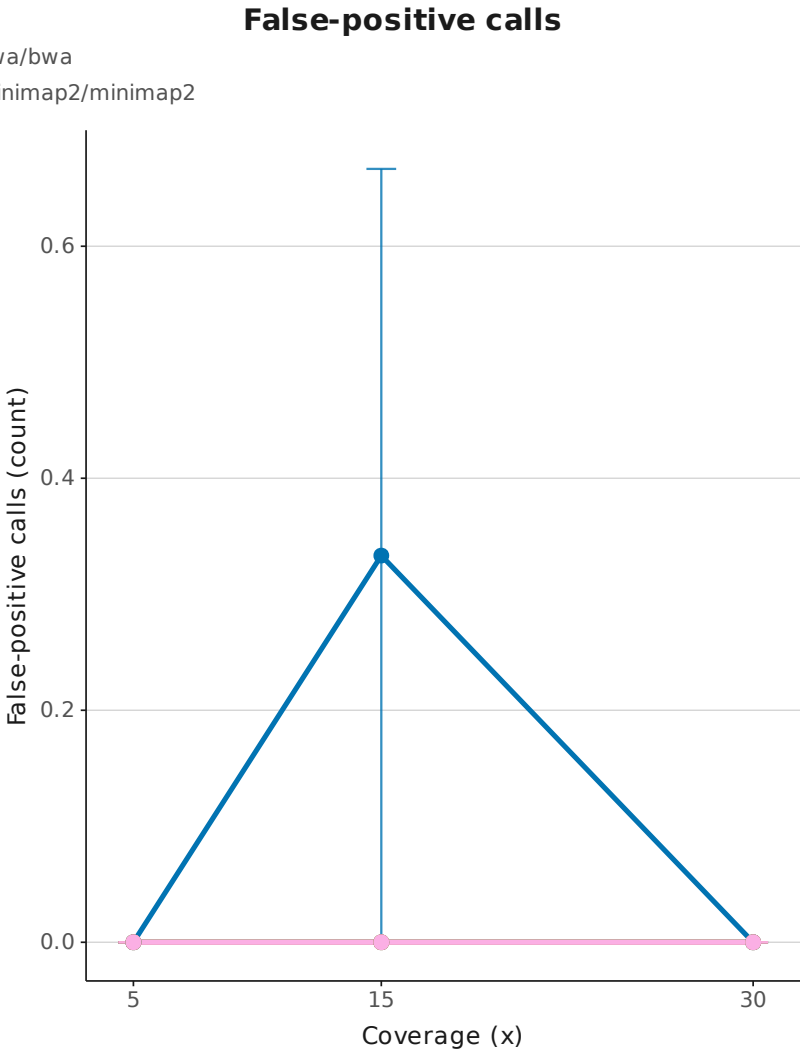
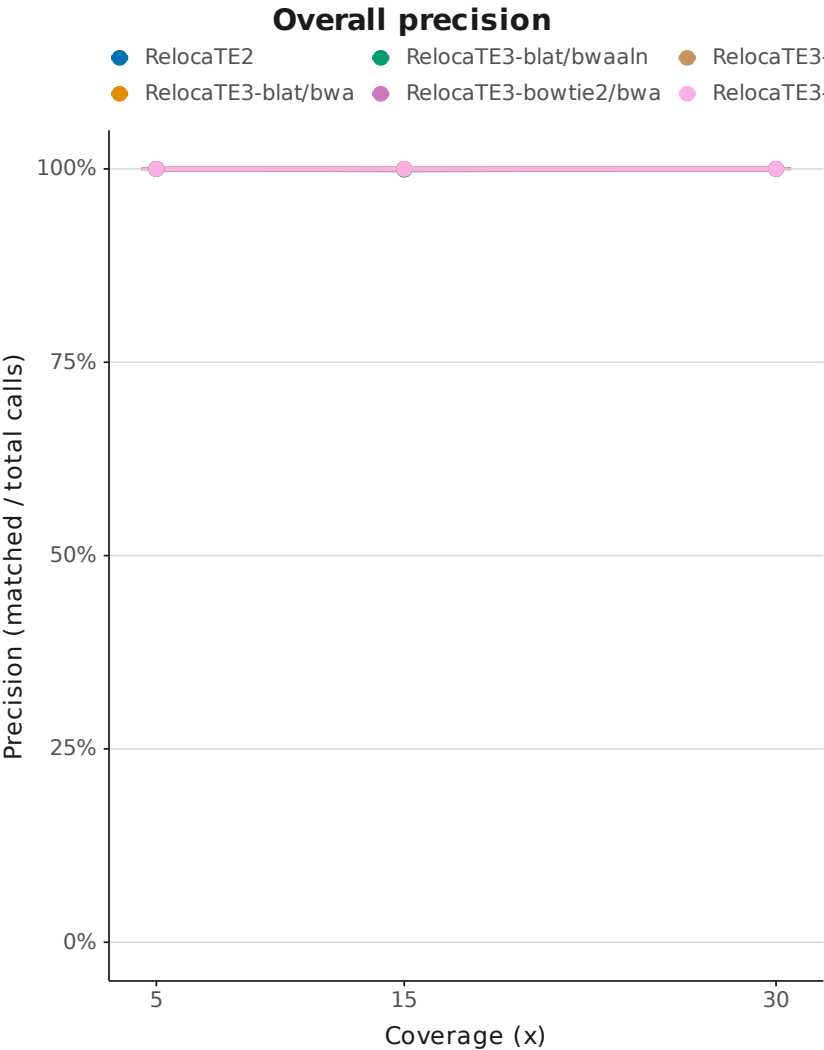
Correct hom/het/somatic call among DETECTED insertions (mean $\pm$ SEM)

- RelocaTE2
- RelocaTE3-blat/bwaaln
- RelocaTE3-bwa/bwa
- RelocaTE3-blat/bwa
- RelocaTE3-bowtie2/bwa
- RelocaTE3-minimap2/minimap2



Precision and false positives vs. coverage

RelocaTE precision = fraction of calls matching a true insertion (mean $\pm$ SEM)



Somatic detection recall by cellular fraction

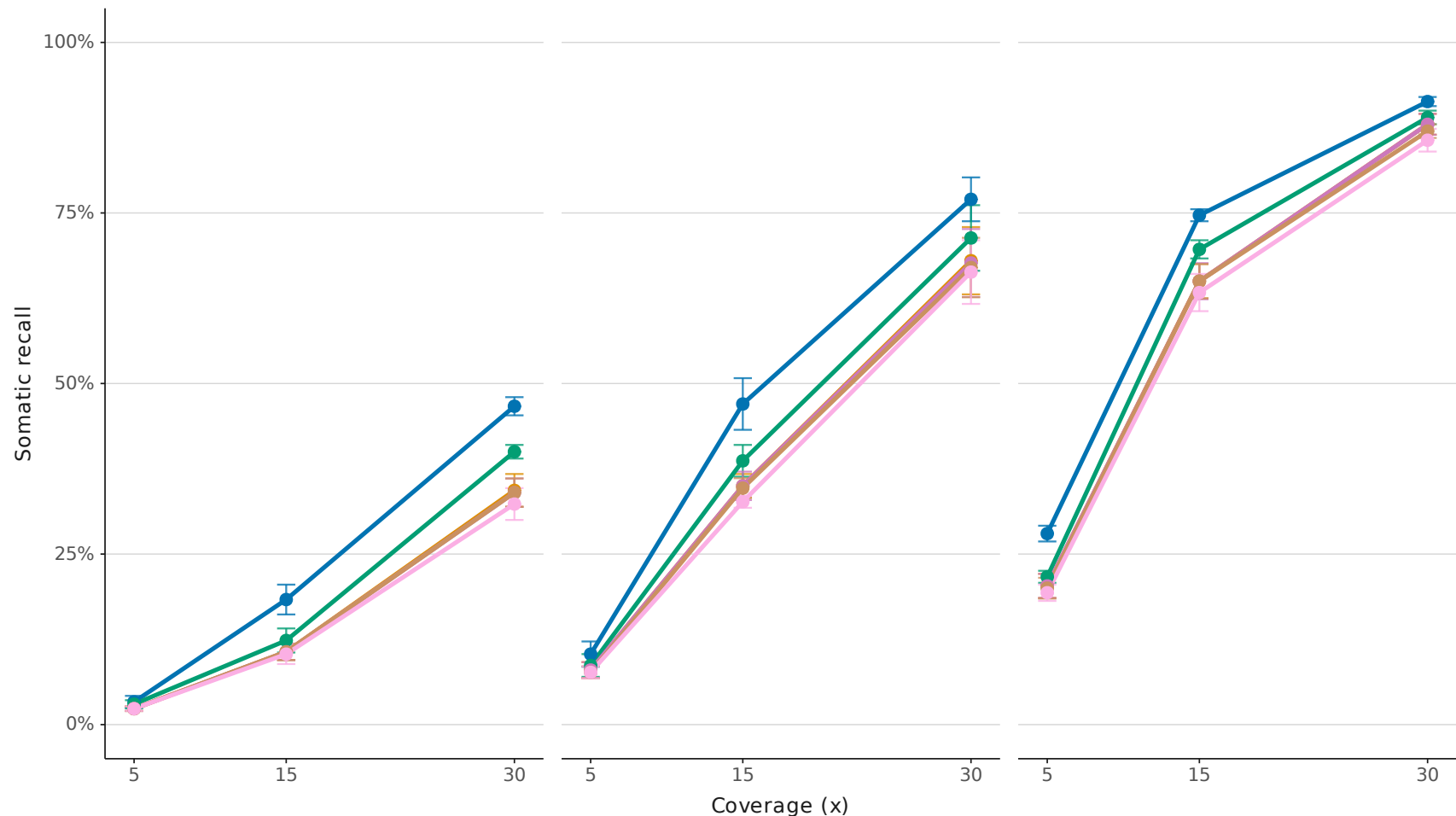
Lower cellular fraction = lower variant allele frequency = harder (mean $\pm$ SEM)

- RelocaTE2
- RelocaTE3-blat/bwaaln
- RelocaTE3-bwa/bwa
- RelocaTE3-blat/bwa
- RelocaTE3-bowtie2/bwa
- RelocaTE3-minimap2/minimap2

Cellular fraction 0.1 (VAF 0.05)

Cellular fraction 0.2 (VAF 0.1)

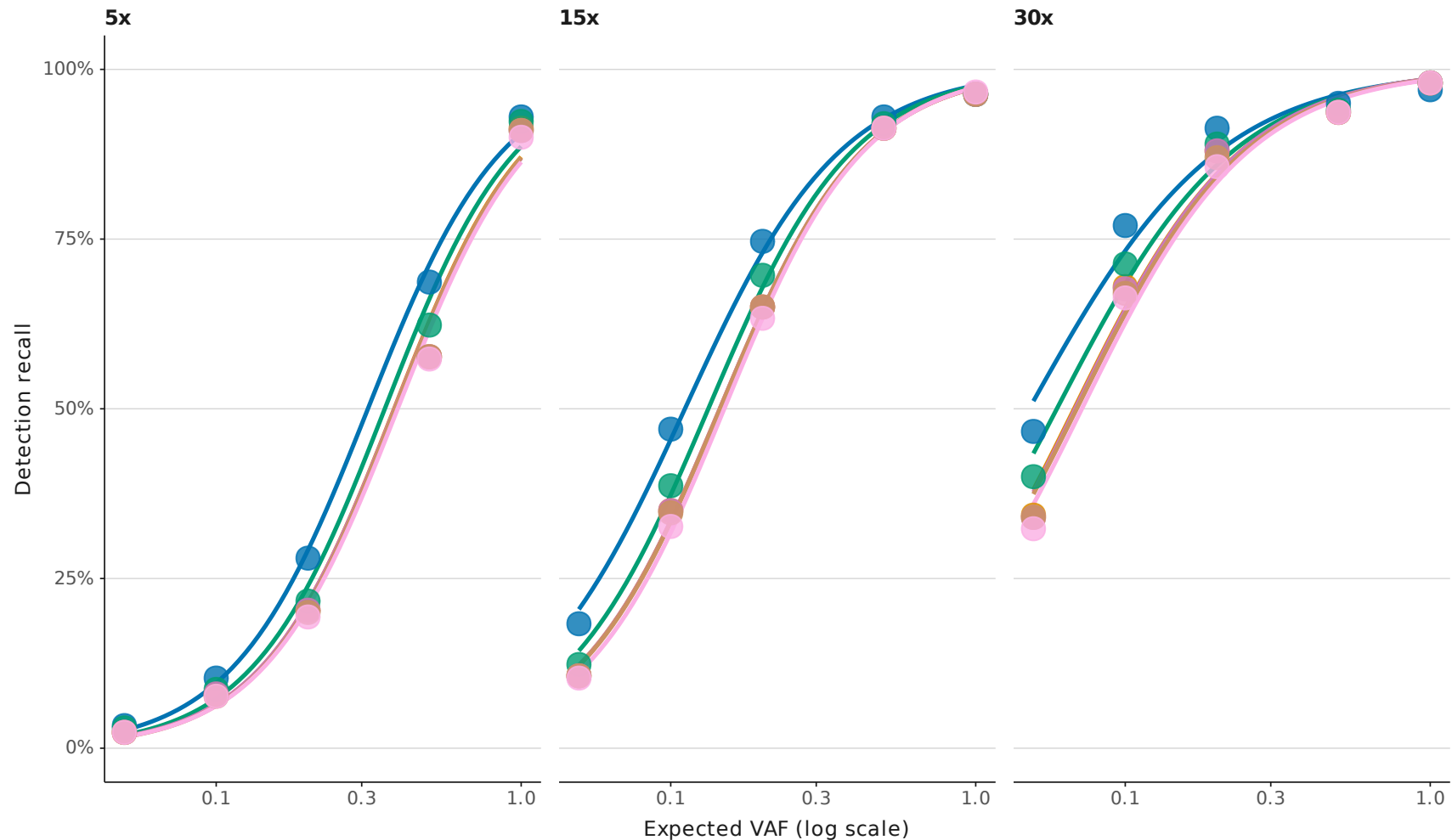
Cellular fraction 0.4 (VAF 0.2)



Detection sensitivity vs variant allele frequency

Logistic fit per caller; VAF at 50% detection = limit of detection (LOD50)

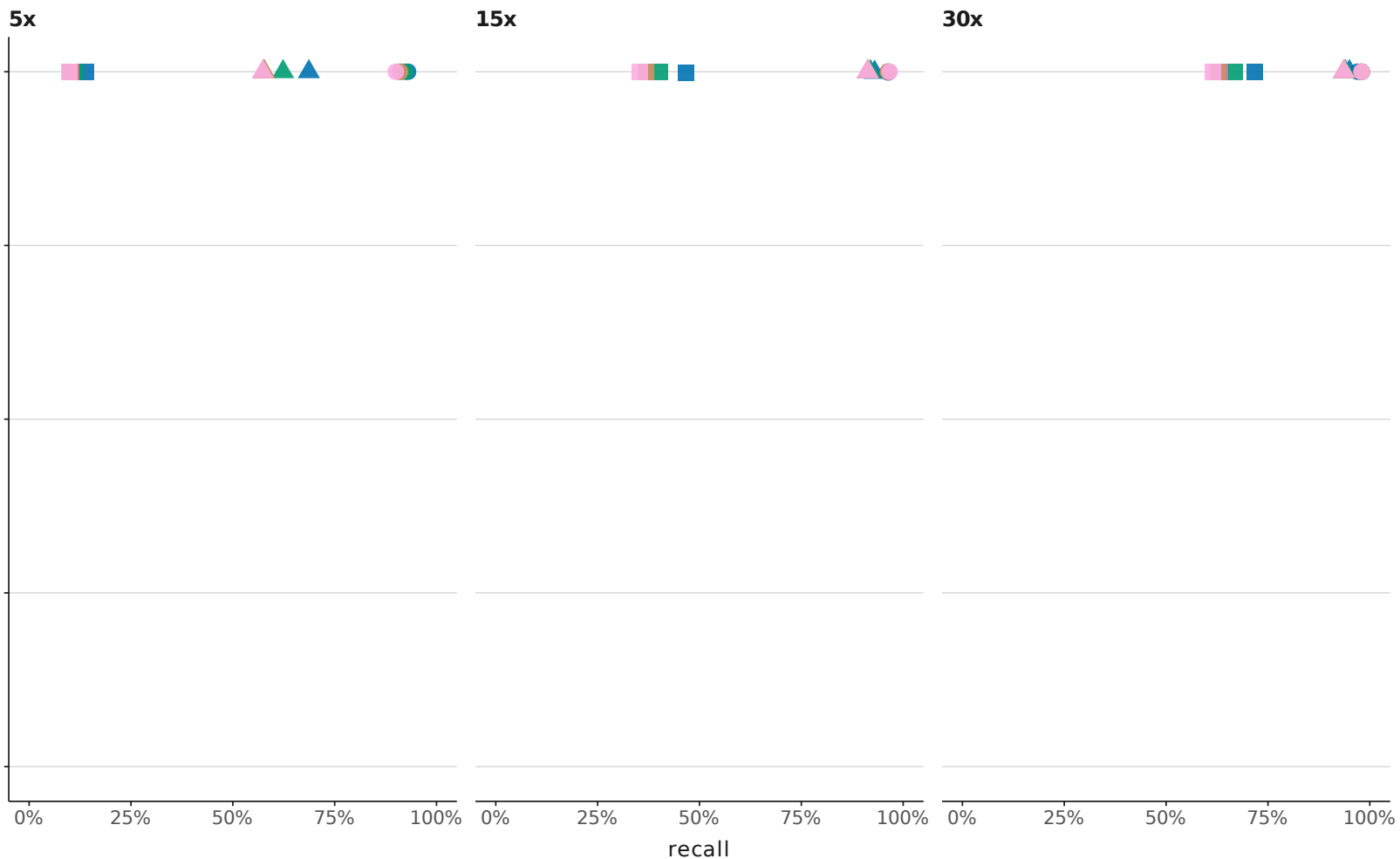
- RelocaTE2
- RelocaTE3-blat/bwaaln
- RelocaTE3-bwa/bwa
- RelocaTE3-blat/bwa
- RelocaTE3-bowtie2/bwa
- RelocaTE3-minimap2/minimap2



Precision-recall operating points

Precision = matched / all calls per sample (global denominator); one point per class

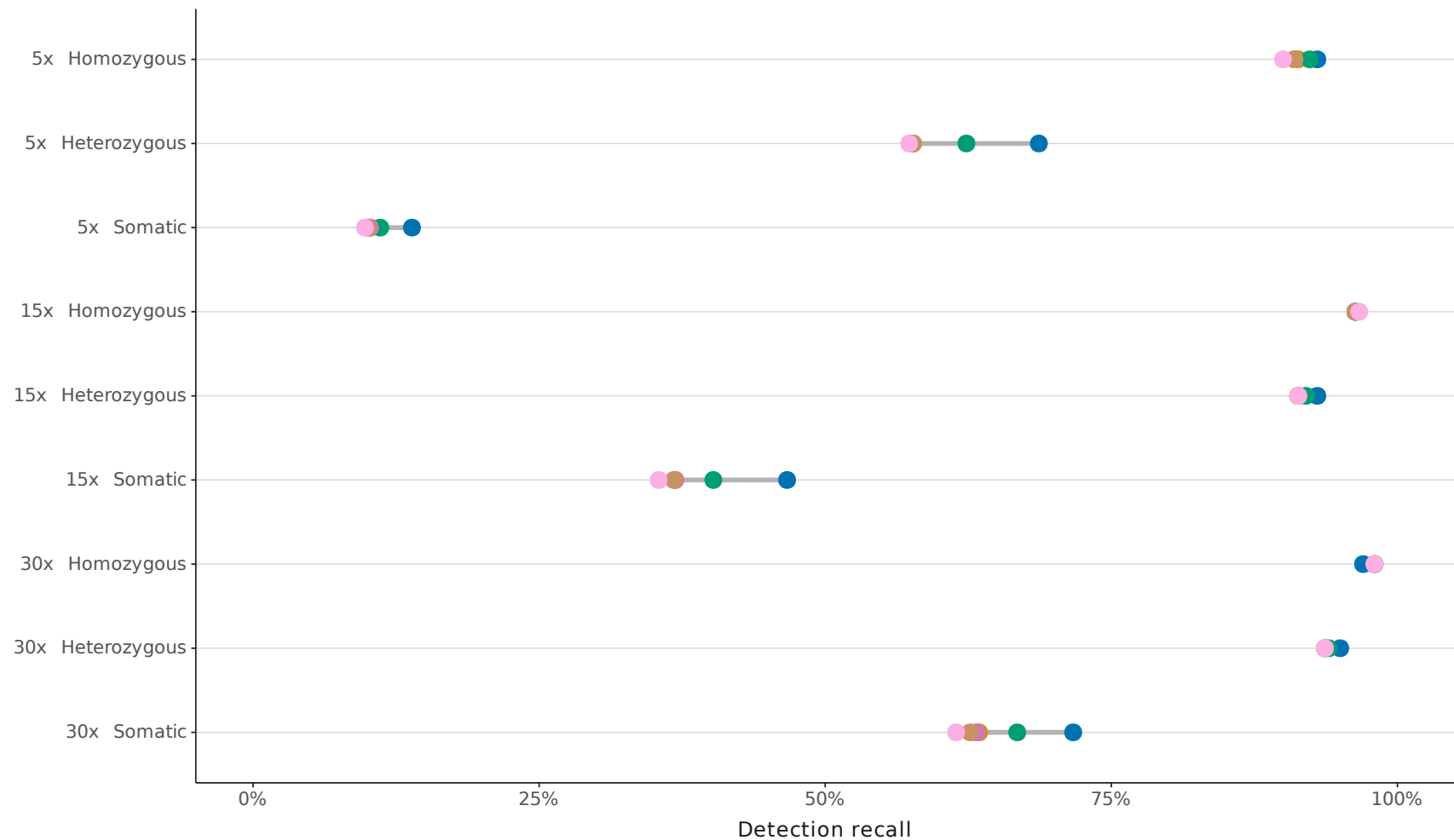
- Homozygous ▲ Heterozygous ■ Somatic
- RelocaTE2 ● RelocaTE3-blat/bwaaln ● RelocaTE3-bwa/bwa
- RelocaTE3-blat/bwa ● RelocaTE3-bowtie2/bwa ● RelocaTE3-minimap2/minimap2



Detection recall by caller

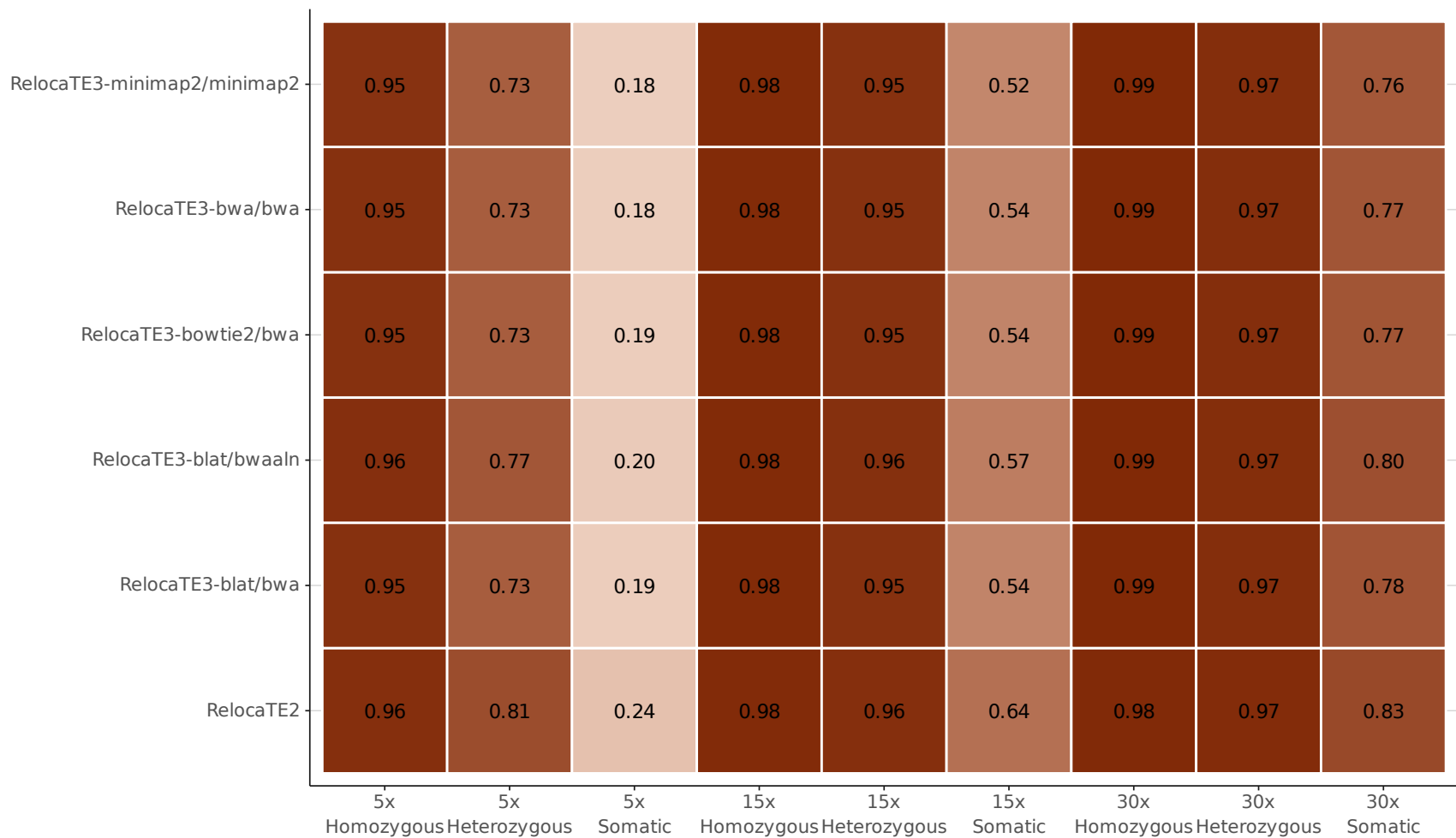
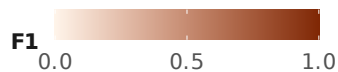
Each row = coverage x class; grey connector spans the 6 callers

- RelocaTE2
- RelocaTE3-blat/bwaaln
- RelocaTE3-bwa/bwa
- RelocaTE3-blat/bwa
- RelocaTE3-bowtie2/bwa
- RelocaTE3-minimap2/minimap2



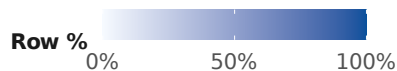
F1 (detection x precision) by condition

Harmonic mean of recall and precision (precision = global-denominator, per sample)



Genotype / detection confusion

Row-normalised: where each truth class's events land (pooled over coverage)



RelocaTE2

Somatic	917 34%	23 1%	1 0%	249 9%	1510 56%
Heterozygous	544 60%	64 7%	162 18%	0 0%	130 14%
Homozygous	2 0%	849 94%	8 1%	0 0%	41 5%

RelocaTE3-blat/bwa

Somatic	747 28%	10 0%	1 0%	237 9%	1705 63%
Heterozygous	560 62%	55 6%	113 13%	0 0%	172 19%
Homozygous	3 0%	844 94%	10 1%	0 0%	43 5%

RelocaTE3-blat/bwaaln

Somatic	806 30%	10 0%	1 0%	246 9%	1637 61%
Heterozygous	549 61%	59 7%	137 15%	0 0%	155 17%
Homozygous	6 1%	847 94%	7 1%	0 0%	40 4%

RelocaTE3-bowtie2/bwa

Somatic	745 28%	10 0%	1 0%	237 9%	1707 63%
Heterozygous	559 62%	56 6%	113 13%	0 0%	172 19%
Homozygous	3 0%	843 94%	10 1%	0 0%	44 5%

RelocaTE3-bwa/bwa

Somatic	738 27%	10 0%	1 0%	236 9%	1715 64%
Heterozygous	560 62%	56 6%	112 12%	0 0%	172 19%
Homozygous	2 0%	844 94%	10 1%	0 0%	44 5%

RelocaTE3-minimap2/minimap2

Somatic	724 27%	9 0%	1 0%	226 8%	1740 64%
Heterozygous	566 63%	58 6%	102 11%	1 0%	173 19%
Homozygous	4 0%	841 93%	9 1%	0 0%	46 5%

Het Hom Hom-exc Somatic Missed

Het Hom Hom-exc Somatic Missed

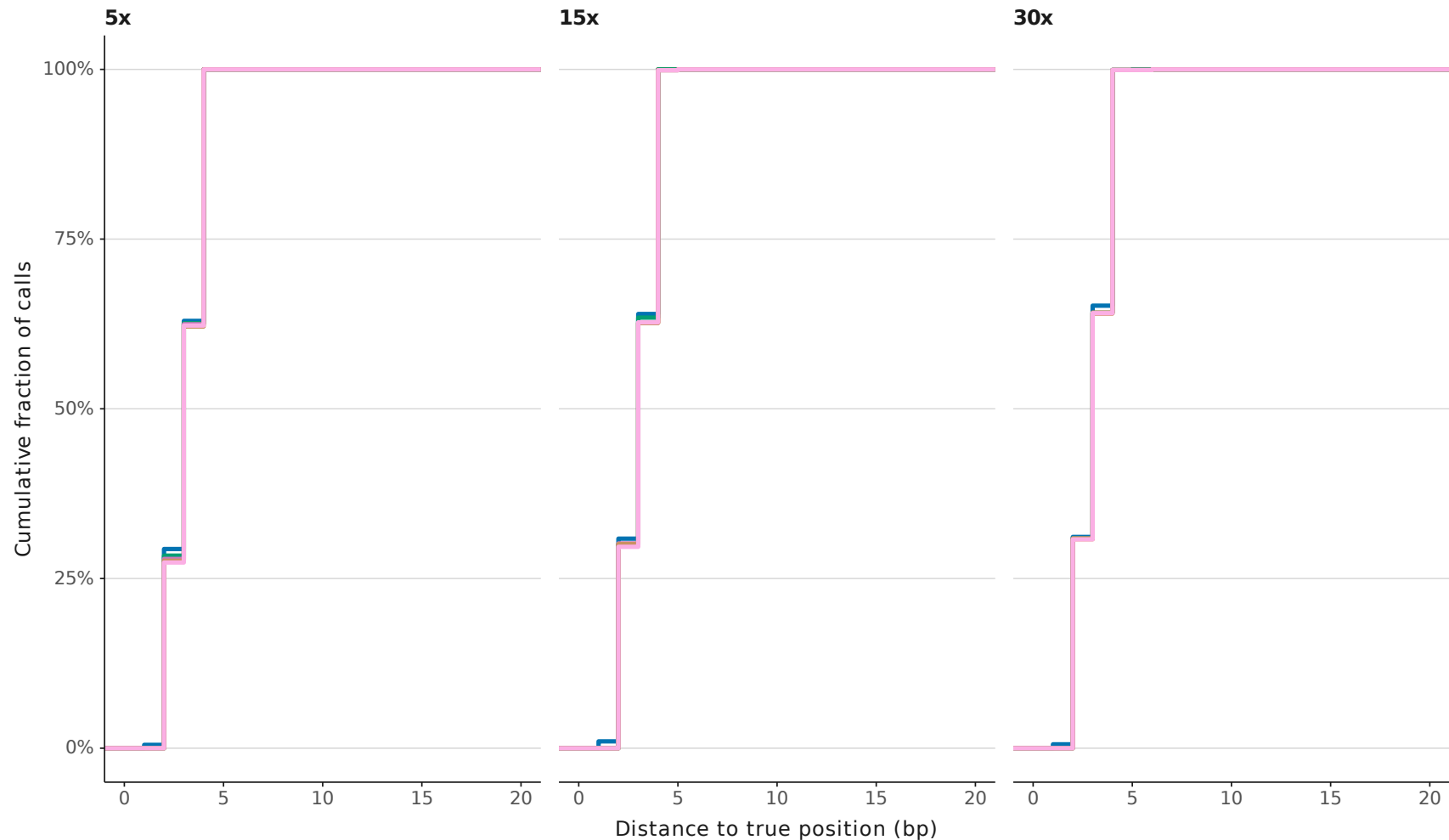
Het Hom Hom-exc Somatic Missed

Called status

Breakpoint positional accuracy

ECDF of $|\text{called} - \text{true}|$ position for matched events; steeper/left = better

- RelocaTE2
- RelocaTE3-blat/bwaaln
- RelocaTE3-bwa/bwa
- RelocaTE3-blat/bwa
- RelocaTE3-bowtie2/bwa
- RelocaTE3-minimap2/minimap2



Per-event caller agreement

Truth events by which caller(s) detected them (pooled over all samples)



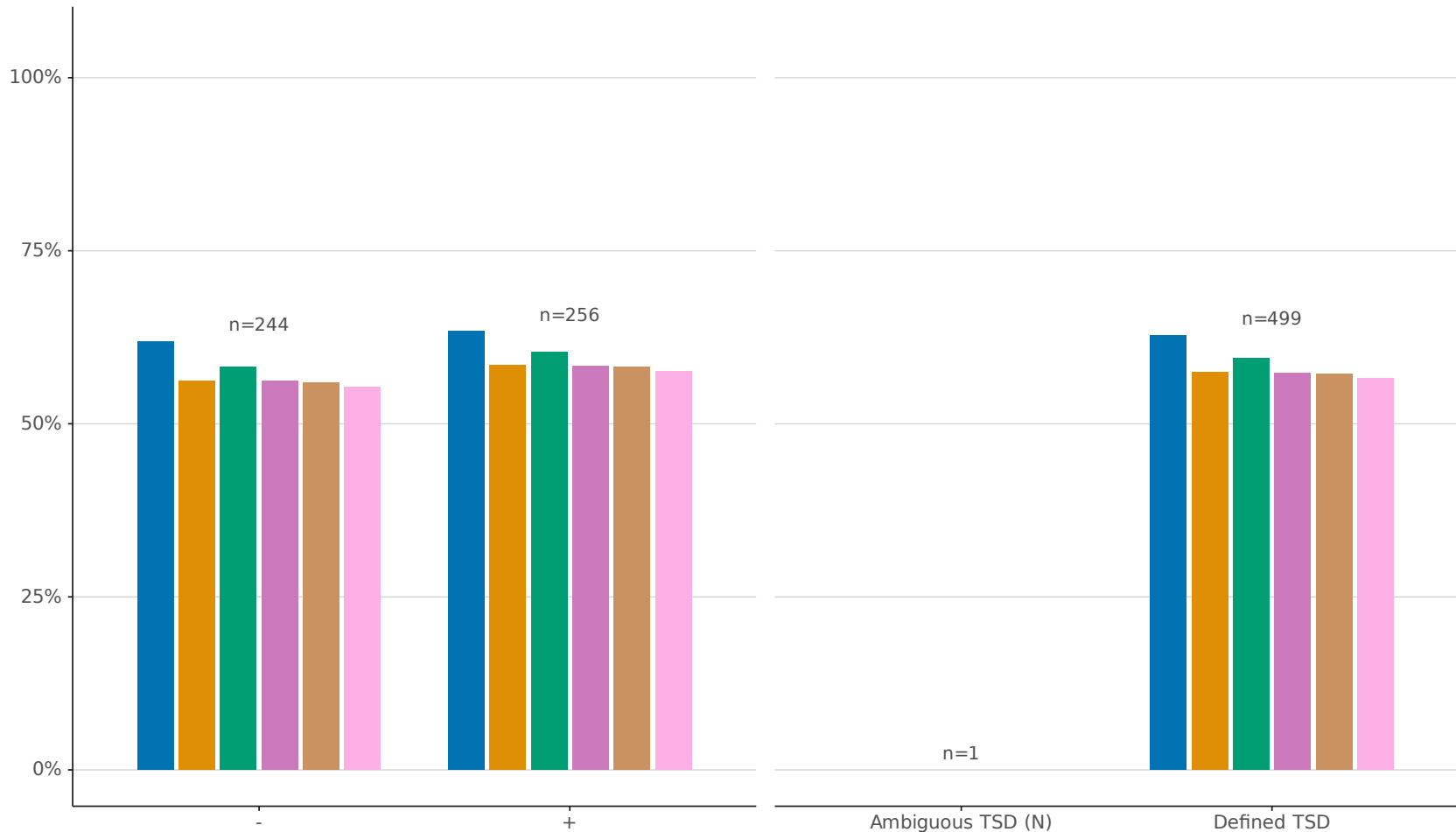
What do missed events have in common?

Detection recall stratified by strand and TSD ambiguity (pooled); n = distinct truth events

RelocaTE2 RelocaTE3-blat/bwaaln RelocaTE3-bwa/bwa
RelocaTE3-blat/bwa RelocaTE3-bowtie2/bwa RelocaTE3-minimap2/minimap2

Strand

TSD definition



Compute resources vs. coverage

Per-sample wall time and peak RSS (mean $\pm$ SEM)

